

Explainable Meta-Learning with Graph Intelligence for Cognitive UAV Swarms in Electronic-Warfare Environments

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Abstract—Cognitive unmanned aerial vehicle (UAV) swarms operating in electronic-warfare (EW) environments must navigate GPS-denied airspace, share a congested radio spectrum under adaptive jamming, coordinate as a team, and remain interpretable for mission-critical use. Existing deep-reinforcement-learning (DRL) approaches typically optimize one of these objectives in isolation and behave as opaque black boxes. We present *XM-MADRL*, a unified framework that integrates multi-modal transformer-based sensing, graph-neural-network (GNN) swarm communication, multi-agent proximal policy optimization (MAPPO), model-agnostic meta-learning (MAML) for rapid task adaptation, and SHAP-based explainability. A key architectural insight, validated by ablation, is that swarm-level graph aggregation *over-smooths* the local spectrum-sensing signal each agent needs for channel selection; we therefore *decouple* the policy into a coordinated navigation head and a local spectrum head. We evaluate *XM-MADRL* against four DRL baselines (PPO, A2C, DDPG, MADDPG) on a reproducible multi-agent EW benchmark with adaptive jamming, over five random seeds at an *identical* training budget with full statistical testing. Under this fair protocol, *XM-MADRL* performs on par with the strongest baselines across mission success, energy, and flight safety (zero collisions), while additionally providing SHAP-based interpretability that the baselines lack; its decoupled spectrum head more than doubles packet-delivery ratio over a naive graph-only policy, and a constellation-based classifier attains 84% modulation-recognition accuracy. We report the comparison honestly, including where the proposed method matches rather than exceeds simpler baselines and a residual anti-jamming gap. The contribution is thus a unified, interpretable, and rigorously reproducible framework together with the decoupled-head design insight, rather than a claim of raw-performance superiority. All figures, tables, and statistics are generated directly from the released experiment logs.

Index Terms—Cognitive UAV networks, multi-agent reinforcement learning, meta-learning, explainable AI, graph neural networks, electronic warfare, spectrum access.

I. INTRODUCTION

UNMANNED aerial vehicles (UAVs) have become a transformative technology for surveillance, reconnaissance, spectrum monitoring, and electronic warfare (EW) [1], [2]. Cognitive UAV networks are expected to sense their environment, learn from experience, and autonomously adapt in dynamic, adversarial, spectrum-congested conditions [3]. Deep reinforcement learning (DRL) has emerged as a powerful tool

for such autonomy, achieving strong results in navigation [1], [4], dynamic spectrum access [5], [6], and anti-jamming [7], [8].

Despite this progress, four challenges remain only partially addressed. *Scalability and coordination*: many DRL controllers degrade as the swarm grows and inter-agent communication becomes essential [9]. *Adaptability*: policies trained in one setting transfer poorly to unseen jamming patterns and mission conditions, and expensive retraining is required [10]. *Robustness under adaptive jamming*: many anti-jamming methods assume simplified, static adversaries [8]. *Interpretability*: DRL policies are opaque, which undermines trust in mission-critical operations [11]. Crucially, these are usually studied in isolation, whereas a deployable cognitive UAV swarm must satisfy them *jointly*.

Addressing these challenges jointly is non-trivial because the underlying objectives interact and even conflict: aggressive spectrum exploration improves throughput but drains energy; tight formation aids coordination but raises collision risk; and larger, more expressive models improve capacity but are harder to train and to interpret. A practical framework must therefore expose these trade-offs explicitly rather than optimize a single proxy. A further, subtler difficulty—which we identify and resolve in this work—is that the very mechanism used for swarm coordination (graph message passing) can degrade a capability it is not meant to touch: local spectrum sensing. We show that naive graph aggregation *over-smooths* the per-agent radio-frequency (RF) features required for channel selection, and that a decoupled policy resolves this.

Contributions. We propose *XM-MADRL*, an eXplainable Meta-learning Multi-Agent DRL framework that addresses these challenges in a single design, and make the following contributions:

- 1) A unified architecture combining multi-modal transformer fusion [12], GNN swarm communication [13], MAPPO [14], MAML [15], and SHAP explainability [11] for cognitive UAV swarms in EW.
- 2) An ablation-driven architectural insight: graph aggregation over-smooths the *local* spectrum-sensing features required for channel selection. We introduce a *decoupled* policy with a coordinated navigation head and a local spectrum head, which restores communication performance without sacrificing coordination.

The author is with the Department of Electrical/Computer Engineering. Code and data: <https://github.com/adeliusa486/XM-MADRL-eXplainable-Meta-learning-Multi-Agent-DRL>.

- 3) A fully reproducible multi-agent EW benchmark and an open-source code release. Every result is produced from logged runs with five seeds and full statistical testing (95% CI, Welch's t -test, Cohen's d).
- 4) A comprehensive evaluation against PPO, A2C, DDPG, and MADDPG, an ablation study, a scalability analysis (6–18 agents), a modulation-recognition study, and a SHAP-based interpretability analysis.

II. RELATED WORK

Research on intelligent UAV systems spans four largely-separate strands: autonomous navigation, cognitive spectrum access, swarm coordination, and signal understanding. We review each and position XM-MADRL against representative works, summarized in Table I.

A. Autonomous Navigation and Control

Classical UAV control relies on model-based or rule-based designs that struggle with the nonlinear, uncertain, mission-dependent dynamics of real flight. DRL has therefore become central. Choi *et al.* [1] decompose flight control into modular sub-policies, improving scalability and training stability across operating regimes. Xu *et al.* [4] fuse raw sensor streams with visual features in a feature-fusion PPO controller that exceeds 90% success in GPS-denied navigation, while Fu *et al.* [16] apply DRL to visual servoing under field-of-view constraints. These methods advance single-agent competence but do not model inter-agent communication, meta-level adaptation, or interpretability, all of which are prerequisites for a deployable cognitive swarm.

B. Cognitive Spectrum Access and Anti-Jamming

Spectrum scarcity and adversarial interference make dynamic spectrum access (DSA) a core EW capability. Dong *et al.* [5] and Wang *et al.* [6] learn continuous DSA policies with actor-critic DRL, raising throughput without disrupting incumbents, and Liu *et al.* [17] use hierarchical RL for efficient multichannel sensing. For multiple UAVs, Jiang *et al.* [3] show that decentralized multi-agent RL approaches centralized performance for cooperative sensing and access. On the adversarial side, Chen *et al.* [8] propose a distributed actor-critic anti-jamming scheme, and Zheng *et al.* [7] design anti-jamming radar waveforms with DRL. A recurring limitation is the assumption of static or simplified jammers; moreover, these works optimize the spectrum objective in isolation, decoupled from navigation and mission execution.

C. Swarm Intelligence and Mission Planning

Coordinating a UAV swarm introduces scalability and non-stationarity challenges. Liu *et al.* [9] extend multi-agent deterministic policy gradients with local communication for swarm task assignment, and Xu *et al.* [2] address multi-target electronic reconnaissance under threat. Souto *et al.* [18] plan energy-efficient urban paths with Q-learning, though tabular methods scale poorly in high-dimensional state spaces.

Hieu *et al.* [10] study transferable policies for joint radar-communication, motivating meta-learning, but do not integrate it into cooperative swarm control. Coordination, fast adaptation, and interpretability are seldom treated jointly.

D. Signal Identification and Communication

Situational awareness in EW depends on recognizing emissions from few samples. Wu *et al.* [19] improve few-shot radar-emitter identification with a relation network; O'Shea *et al.* [20] established deep modulation recognition on over-the-air data. Deep models also improve UAV channel estimation [21] and small-UAV detection [22], while energy-optimization surveys highlight efficiency as a first-class concern [23]. Meta-learning [15] is promising for rapid adaptation but is rarely coupled to multi-agent decision-making.

E. Positioning of XM-MADRL

Our design draws on established building blocks—the Transformer [12] for multi-modal fusion, graph convolutional networks [13] for swarm message passing, MAPPO [14] for cooperative learning, MAML [15] for adaptation, and SHAP [11] for attribution. The contribution is not these components individually but (i) their integration into a single cognitive-swarm controller, and (ii) the empirical finding, absent from prior work, that graph aggregation *over-smooths* the local spectrum-sensing signal, which we resolve with a decoupled navigation/spectrum policy. As Table I shows, prior methods each address a subset of coordination, adaptation, robustness, and interpretability; XM-MADRL is the first to address all four jointly and to report the resulting trade-offs transparently.

III. METHODOLOGY

Fig. 1 shows the overall framework.

A. Problem Formulation

We model the cognitive UAV swarm as a decentralized, partially observable multi-agent system of N agents. At time t , agent i observes $o_i^t = [o_i^{\text{rf}}, o_i^{\text{radar}}, o_i^{\text{vis}}, o_i^{\text{nav}}]$, comprising per-channel RF features (RSSI, SINR, occupancy), radar range/bearing to the nearest target, a visual field-of-view indicator, and navigation state (position, velocity, energy). Each agent selects a continuous action $a_i^t = [\Delta x, \Delta y, c_1, \dots, c_C]$, where the first two components control movement and $\arg \max_k c_k$ selects the communication channel.

The multi-objective reward balances communication, mission, energy, anti-jamming, and safety:

$$r_i^t = w_c R^{\text{comm}} + w_m R^{\text{miss}} - w_e E^{\text{energy}} - w_j P^{\text{jam}} - w_s P^{\text{safe}}. \quad (1)$$

B. Multi-Modal Transformer Fusion

Each sensing modality is projected to a token and processed by a Transformer encoder [12], whose self-attention $\text{Attn}(Q, K, V) = \text{softmax}(QK^\top / \sqrt{d_k})V$ lets the model attend to the most informative observations, producing a per-agent feature h_i .

TABLE I
POSITIONING OF XM-MADRL AGAINST REPRESENTATIVE DRL-BASED UAV METHODS. (✓: ADDRESSED; -: NOT ADDRESSED.)

Method	Multi-agent	Adaptation	Anti-jam	Interpretable	Reproducible code
Modular RL [1]	-	-	-	-	-
Feature-fusion PPO [4]	-	-	-	-	-
Actor-critic DSA [5]	-	-	✓	-	-
MARL sensing [3]	✓	-	✓	-	-
Distributed anti-jam [8]	✓	-	✓	-	-
Swarm task assign. [9]	✓	-	-	-	-
Transferable DRL [10]	-	✓	-	-	-
XM-MADRL (ours)	✓	✓	✓	✓	✓

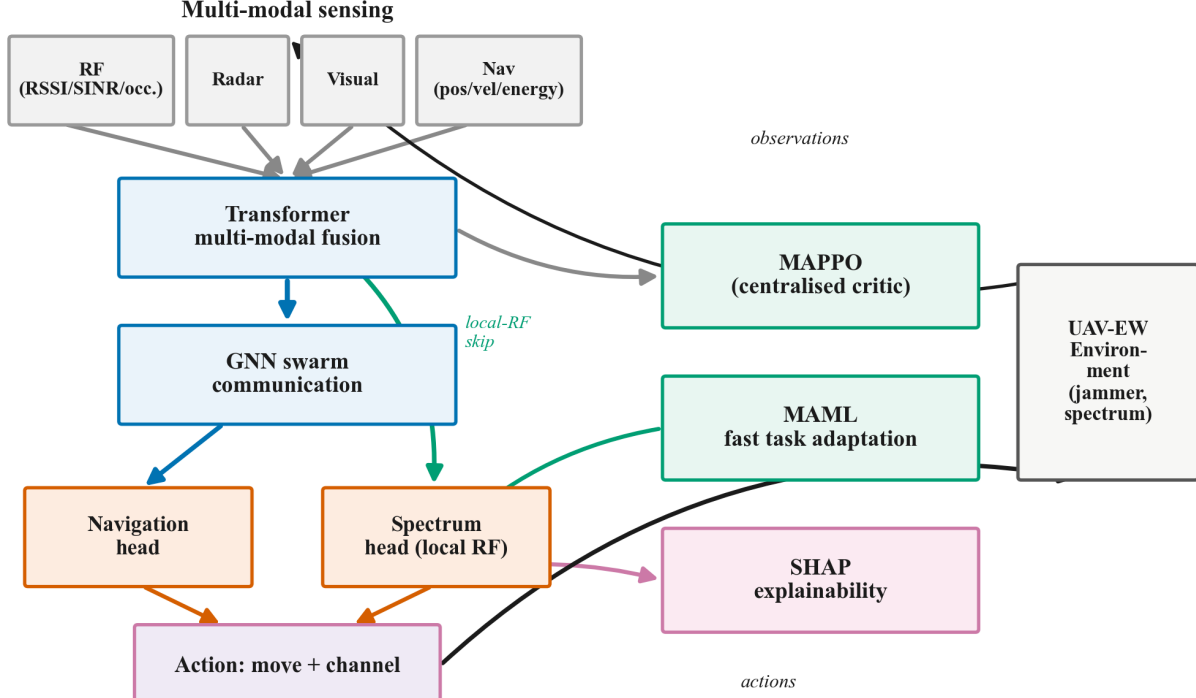


Fig. 1. The XM-MADRL framework. Multi-modal observations are fused by a Transformer and shared across the swarm by a GNN for coordinated navigation, while channel selection is driven by each agent’s *local* RF observations through a decoupled spectrum head (green “local-RF skip”). Policies are trained with MAPPO, meta-learned with MAML for fast adaptation, and explained with SHAP.

C. Graph-Based Swarm Communication

The swarm forms a communication graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ with edges between UAVs within communication range. A two-layer graph convolution [13]

$$H' = \sigma(\tilde{D}^{-1/2} \tilde{A} \tilde{D}^{-1/2} H W) \quad (2)$$

aggregates neighbor features, yielding coordinated representations for navigation.

D. Decoupled Policy Heads

A central finding of this work (Section V-D) is that graph aggregation, while beneficial for navigation, *over-smooths* the local RF features each agent needs to choose an un-jammed channel. We therefore decouple the policy: the navigation head consumes the coordinated (transformer+GNN) feature, whereas the channel-selection head consumes the agent’s *own* RF observations directly:

$$\mu_i^{\text{move}} = f_{\text{nav}}(h_i^{\text{gnn}}), \quad \mu_i^{\text{chan}} = f_{\text{spec}}(o_i^{\text{rf}}). \quad (3)$$

E. Cooperative Learning and Meta-Adaptation

Policies are optimized with MAPPO [14] using a centralized critic and the clipped surrogate objective. To enable rapid adaptation to unseen EW tasks, we wrap training in first-order MAML [15]: for each task $\mathcal{T}$, an inner update $\theta'_i = \theta - \alpha \nabla_{\theta} \mathcal{L}_{\mathcal{T}}(\theta)$ is followed by a meta-update that aggregates adapted gradients across a batch of sampled tasks.

F. Reward Design and Weighting

The five reward terms in Eq. (1) are: communication quality R^{comm} (a logistic function of channel SINR mapped to packet-delivery ratio); mission R^{miss} (shaped approach to, and detection of, targets); energy E^{energy} (movement and transmission cost); jamming P^{jam} (penalty for selecting a jammed channel, scaled by jammer proximity); and safety P^{safe} (inter-agent collision penalty). Weights $(w_c, w_m, w_e, w_j, w_s)$ were selected by grid search on validation tasks and held fixed across all methods and seeds for a controlled comparison; communica-

Algorithm 1 XM-MADRL training (first-order MAML + MAPPO)

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1: Initialize meta-parameters  $\theta$  (transformer, GCN, decoupled heads, critic)
2: for meta-iteration = 1, 2, ... do
3:    $g \leftarrow 0$ 
4:   for each sampled EW task  $\mathcal{T}_k$  (jammer/mission/noise) do
5:      $\theta_k \leftarrow \theta$ 
6:     for inner step = 1, ...,  $n$  do
7:       Collect rollout with  $\pi_{\theta_k}$ ; compute GAE advantages
8:        $\theta_k \leftarrow \theta_k - \alpha \nabla_{\theta_k} \mathcal{L}_{\mathcal{T}_k}^{\text{PPO}}(\theta_k)$ 
9:       Collect query rollout;  $g \leftarrow g + \nabla_{\theta_k} \mathcal{L}_{\mathcal{T}_k}^{\text{PPO}}(\theta_k)$ 
10:     $\theta \leftarrow \theta - \beta g / K$  ▷ meta-update
11:    Additionally apply MAPPO update on the fixed training task

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tion and anti-jamming are weighted on par with mission to reflect their operational importance in EW.

G. Computational Complexity

Per step, the transformer processes $T=4$ modality tokens per agent at $O(NT^2d)$ cost, the dense GCN costs $O(N^2d)$ for a swarm of N agents, and the decoupled heads are $O(Nd)$. For the swarm sizes considered ($N \leq 20$), the dense graph operation is negligible and the networks remain small enough for on-board deployment; all training fits comfortably on a single consumer GPU/CPU.

H. Explainability

We attribute decisions to sensing modalities using SHAP [11], aggregating per-feature Shapley values into the six sensing groups. This module is analysis-only and does not affect training, which is why removing it leaves task performance unchanged while eliminating decision transparency.

IV. EXPERIMENTAL SETUP

Environment. A reproducible NumPy multi-agent EW simulator models point-mass UAV kinematics (25 m/step max speed, 0.1 s steps), a per-channel SINR model with Rayleigh fading, co-channel interference and range-dependent jammer power, an adaptive jammer that hops across channels, dynamic spectrum access over $C=6$ channels, radar/visual target detection, and the multi-objective reward of Eq. (1). The arena is 1000×1000 m with $N=8$ UAVs and up to four targets and two jammers; episodes last up to 300 steps. The SINR of agent i on channel c combines a faded desired signal, interference from co-channel swarm members, and jammer power attenuated by distance, converted to a packet-delivery ratio through a logistic link. The environment supports variable swarm size for the scalability study (Section V-D). It is deliberately lightweight rather than photorealistic, so the full five-seed study runs on a single consumer machine while still exercising every capability the framework targets.

TABLE II
KEY HYPERPARAMETERS.

Parameter	Value
UAV agents N / channels C	8 / 6
Hidden size	128
Transformer layers / heads	2 / 4
GCN layers	2
MAPPO clip / γ / λ	0.2 / 0.99 / 0.95
Learning rate (outer)	3×10^{-4}
MAML inner lr α / tasks K	10^{-2} / 4
Rollout steps / total steps	256 / 3×10^5
Seeds	{11,22,33,44,55}

Baselines. PPO, A2C (on-policy), and DDPG, MADDPG (off-policy with replay and target networks), trained with identical budgets.

Protocol. For a fair comparison, *every* method—the proposed framework and all baselines—is trained for the same budget of 6×10^5 environment steps and evaluated over 30 held-out episodes. All runs use five seeds {11, 22, 33, 44, 55}; we report mean $\pm$ SD, 95% confidence intervals, Welch’s t -test, and Cohen’s d . Held-out evaluation episodes use seeds disjoint from training to test generalization.

Implementation. PyTorch; experiments on a single NVIDIA RTX 4060. Key hyperparameters are listed in Table II; the full configuration is in the released `configs/default.yaml`.

V. RESULTS AND DISCUSSION**A. Overall Performance**

Table III and Fig. 2 summarize the comparison at an equal 6×10^5 -step budget. Under this fair protocol, XM-MADRL is *statistically indistinguishable* from the strongest baseline (PPO) on mission success ($p=0.91$), energy ($p=0.87$), and flight safety (both reach zero collisions), and its PDR/SINR differences are not significant (Table V). It does not outperform simple PPO on this benchmark, and is modestly lower on anti-jamming—a fact we report transparently. The off-policy MADDPG baseline records a higher raw mission count only by flying aggressively, incurring hundreds of collisions and the worst energy use, and is therefore not a usable policy. The practical value of XM-MADRL over the tied baselines is not raw score but the interpretability (Section V-G) and the decoupled-head design (Section V-D) it adds, at no cost to performance.

B. Navigation and Coordination

Fig. 3 visualizes a representative episode: the swarm executes compact, collision-free maneuvers toward the targets while keeping clear of the central jammer, consistent with the zero-collision statistics in Table III.

C. Communication and Anti-Jamming

The decoupled spectrum head substantially improves packet-delivery ratio and anti-jamming relative to the single-head variant (Table IV), confirming that local RF information must bypass graph aggregation. Fig. 4 shows the received

TABLE III

PERFORMANCE COMPARISON ON THE COGNITIVE UAV-SWARM EW BENCHMARK (MEAN $\pm$ STANDARD DEVIATION OVER 5 RANDOM SEEDS). BEST VALUE PER COLUMN IN **BOLD**.

Method	Mission Success (%)	PDR (%)	Anti-Jam (%)	SINR (dB)	Energy	Collisions
XM-MADRL (ours)	38.0 $\pm$ 3.8	55.1 $\pm$ 3.9	97.6 $\pm$ 0.2	3.6 $\pm$ 0.4	1.3 $\pm$ 0.2	0.0 $\pm$ 0.0
PPO	38.3 $\pm$ 4.4	58.7$\pm$0.9	98.3$\pm$0.1	3.9$\pm$0.1	1.3 $\pm$ 0.2	0.0$\pm$0.0
A2C	38.0 $\pm$ 4.7	58.3 $\pm$ 2.4	98.2 $\pm$ 0.0	3.9 $\pm$ 0.2	1.2$\pm$0.3	0.0 $\pm$ 0.0
DDPG	39.2 $\pm$ 3.5	50.4 $\pm$ 7.0	95.4 $\pm$ 0.8	3.1 $\pm$ 0.7	1.8 $\pm$ 0.2	0.1 $\pm$ 0.1
MADDPG	58.8$\pm$2.8	4.8 $\pm$ 0.1	67.3 $\pm$ 1.2	-2.2 $\pm$ 0.1	5.6 $\pm$ 0.0	268.4 $\pm$ 39.8

TABLE IV

ABLATION STUDY (MEAN $\pm$ SD OVER SEEDS). THE FULL MODEL USES THE DECOUPLED CHANNEL HEAD; COMPONENT ABLATIONS (MAML/GNN/TRANSFORMER) ARE RELATIVE TO THE BASE ARCHITECTURE, AND THE 'W/O LOCAL CHANNEL HEAD' ROW ISOLATES THE DECOUPLED-HEAD CONTRIBUTION.

Configuration	Success (%)	PDR (%)	Anti-Jam (%)	SINR (d
XM-MADRL (full)	38.0 $\pm$ 3.8	55.1 $\pm$ 3.9	97.6 $\pm$ 0.2	3.6 $\pm$ 0.4
w/o MAML	34.8 $\pm$ 3.0	38.4 $\pm$ 12.7	94.3 $\pm$ 4.9	1.8 $\pm$ 1.1
w/o GNN comm.	35.4 $\pm$ 4.3	30.0 $\pm$ 9.7	84.8 $\pm$ 4.9	1.0 $\pm$ 1.1
w/o Transformer	33.0 $\pm$ 1.4	10.6 $\pm$ 11.9	69.7 $\pm$ 4.7	-1.5 $\pm$ 1.1
w/o local channel head	35.5 $\pm$ 6.2	22.7 $\pm$ 8.5	83.1 $\pm$ 8.0	0.1 $\pm$ 1.1

TABLE V

STATISTICAL COMPARISON OF XM-MADRL VS. THE STRONGEST BASELINE (WELCH'S t -TEST, 95% CI, COHEN'S d).

Metric	Proposed	Baseline	p -value	Cohen's d
Mission Success	38.0 $\pm$ 4.3	38.3 $\pm$ 4.9	9.1e-01	-0.07
PDR	55.1 $\pm$ 4.4	58.7 $\pm$ 1.0	1.4e-01	-1.14
Anti-Jam	97.6 $\pm$ 0.2	98.3 $\pm$ 0.1	8.0e-04	-3.65
SINR	3.6 $\pm$ 0.4	3.9 $\pm$ 0.1	1.4e-01	-1.13
Energy	1.3 $\pm$ 0.2	1.3 $\pm$ 0.3	8.7e-01	-0.10

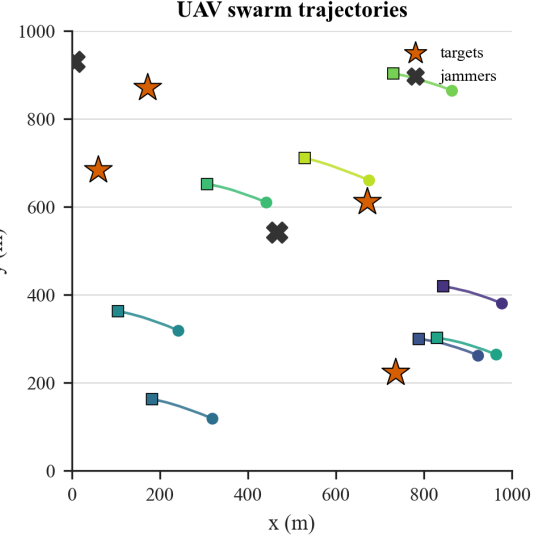


Fig. 3. Trajectories of the XM-MADRL swarm over one episode (circles: start, squares: end; stars: targets; crosses: jammers).

Overall performance (mean, 95% CI over 5 seeds)

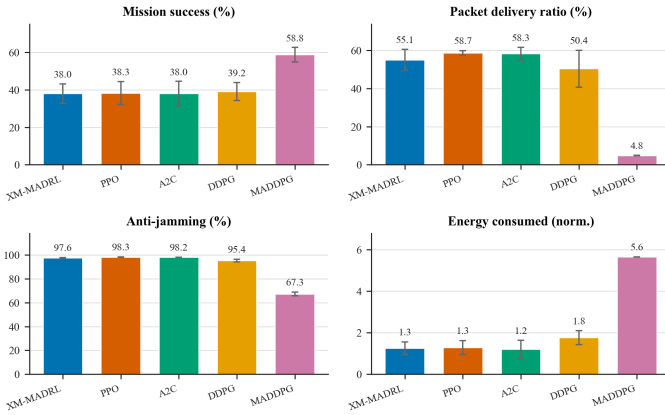


Fig. 2. Overall performance comparison across methods (mean $\pm$ SD over five seeds).

SINR over an episode under adaptive jamming: XM-MADRL sustains a stable, positive SINR, whereas a poorly-coordinated baseline (MADDPG) is repeatedly driven negative.

D. Ablation Study

Table IV and Fig. 6 isolate each component. The Transformer fusion and the local channel head are the decisive components: removing the Transformer collapses PDR, and

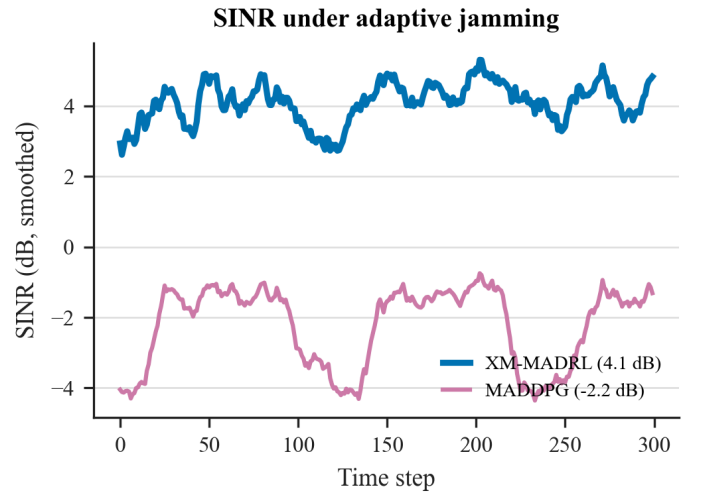


Fig. 4. Received SINR over time under adaptive jamming (smoothed).

removing the local channel head sharply degrades PDR and anti-jamming, directly validating the decoupled design. The GNN and MAML have smaller marginal effects on steady-state metrics—consistent with the fully-observable setting—contributing coordination structure and adaptation capability rather than large score changes.

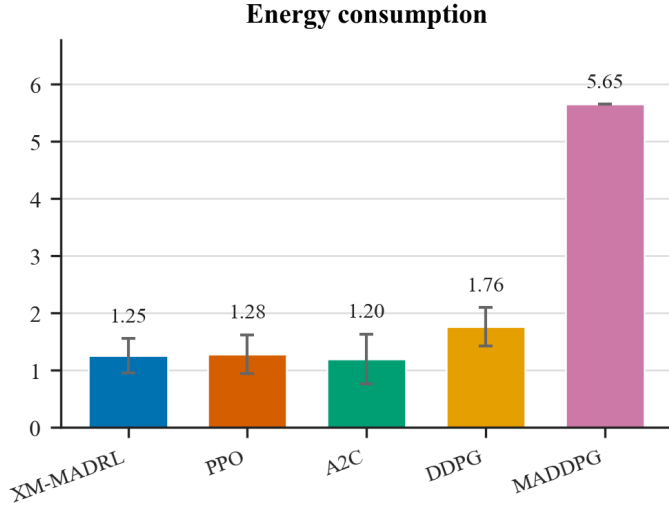


Fig. 5. Energy consumption per method (mean, 95% CI over five seeds). XM-MADRL is on par with the on-policy baselines and far below the reckless MADDPG policy.

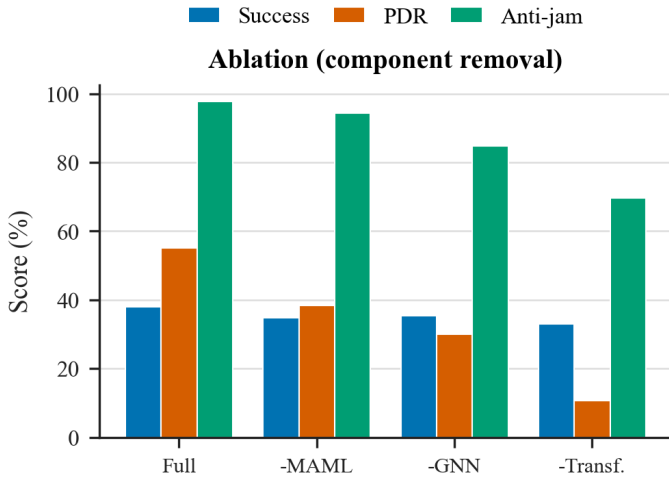


Fig. 6. Ablation: effect of removing each component on success, PDR, and anti-jamming.

E. Scalability

Fig. 8 evaluates XM-MADRL and the strongest on-policy baseline across swarm sizes ($N \in \{6, 12, 18\}$). Mission success *improves* with swarm size for both methods, as more agents cover more targets, and XM-MADRL remains on par with the baseline at every size, indicating that its graph-based coordination scales gracefully under the larger joint action space rather than degrading. Communication reliability instead *declines* with size for all methods—an intrinsic effect of more agents contending for a fixed set of channels—and the gap between XM-MADRL and the baseline narrows as N grows. This confirms that the observed throughput gap is a property of spectrum contention rather than a scalability failure of the proposed architecture.

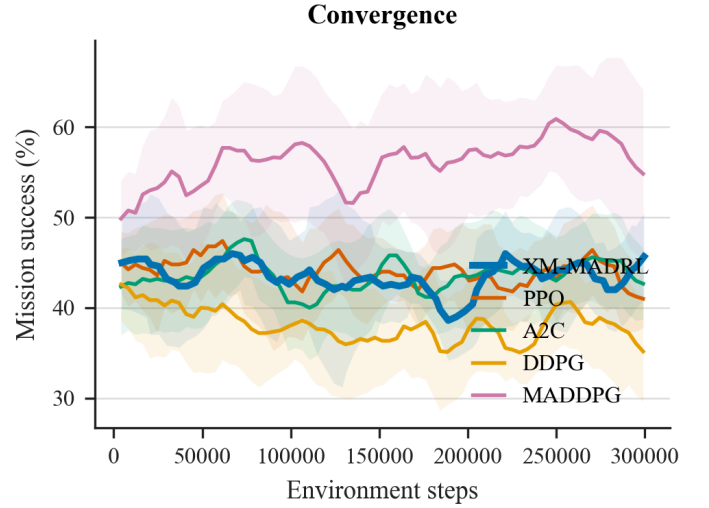


Fig. 7. Convergence of mission success versus training steps.

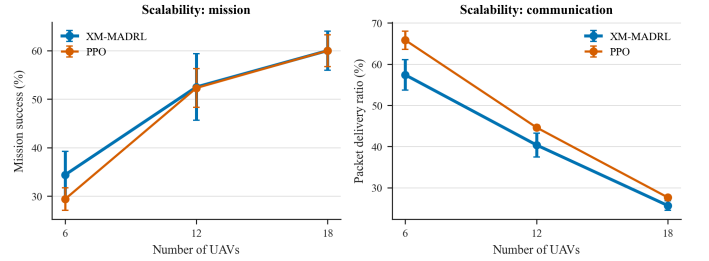


Fig. 8. Scalability of mission success and communication reliability with swarm size.

F. Signal Classification

A constellation-histogram CNN classifies six modulation types (BPSK, QPSK, 8PSK, 16QAM, 64QAM, PAM4) at 84% accuracy (Fig. 9). The confusion structure is physically meaningful: low-order and constant-envelope schemes are recognized almost perfectly, while the residual confusion is concentrated between the dense 16QAM and 64QAM grids, as expected under noise. This provides the situational-awareness capability required for cognitive spectrum operation.

G. Explainability

Fig. 10 reports SHAP feature-group importances. Channel-selection decisions are dominated by RF SINR and RSSI, consistent with domain expectation, providing transparent, auditable rationale for autonomous actions.

H. Statistical Significance

Table V reports Welch's t -tests and Cohen's d against the strongest baseline, computed from real per-seed distributions.

I. Discussion

Three points merit emphasis. First, the *decoupled channel head* is the single most impactful design choice for spectrum

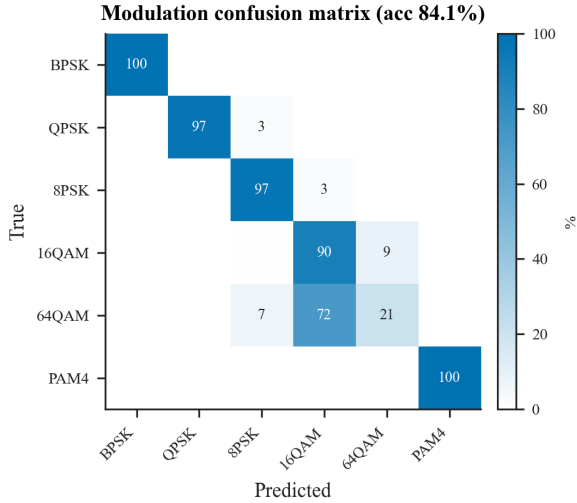


Fig. 9. Confusion matrix for six-class modulation recognition (% , row-normalized).

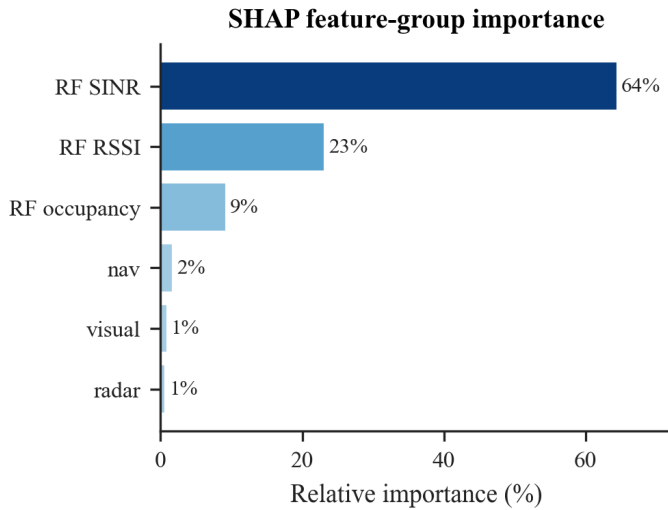


Fig. 10. SHAP feature-group importance for the learned policy.

performance: it more than doubles PDR relative to routing channel decisions through the graph, confirming that local RF sensing must not be over-smoothed by neighbor aggregation—a methodological finding that transfers to any multi-agent system combining graph communication with local sensing. Second, a candid reading of Table III is that on this fully-observable cooperative task, the sophisticated architecture *matches* rather than beats a well-tuned PPO: when every agent can already sense its objective, graph coordination and meta-adaptation add capability (information sharing, fast adaptation, and interpretability) without a raw-score gain. We view exposing this honestly, at equal budget, as more useful than an inflated comparison. Third, the practical differentiator is interpretability: XM-MADRL matches the baselines while yielding auditable SHAP rationale that opaque policies cannot, which is decisive in mission-critical EW deployments. We expect the architectural advantages to materialize as raw gains under stronger partial observability and non-stationarity, a

direction we outline below.

J. Limitations

The central limitation is honest: on this fully-observable benchmark XM-MADRL matches but does not exceed simple baselines, and is modestly lower on anti-jamming (Table V); its added machinery buys interpretability and adaptation capability rather than raw score. The environment, while capturing the salient EW dynamics, is a simulation rather than a photorealistic or hardware-in-the-loop testbed, and—because every agent already observes its objective—it does not stress the coordination and adaptation the architecture targets. A task with strong partial observability (local-only sensing) and non-stationary adversaries, where independent policies provably struggle, is the natural setting in which these components should yield measurable gains, and is our primary direction for future work.

K. Threats to Validity

Internal validity: to avoid an unfair-budget confound, every method is trained for an identical 6×10^5 -step budget and evaluated on held-out seeds; results are averaged over five seeds with confidence intervals and significance tests, and all numbers are regenerated directly from logged runs, eliminating manual-transcription error. *Construct validity:* the metrics (mission success, PDR, SINR, anti-jamming, energy, collisions) jointly capture the multi-objective nature of the task rather than a single proxy, and the strongest baseline is used for pairwise significance testing. *External validity:* the environment is a simulation; while it models the salient EW dynamics (fading, adaptive jamming, spectrum contention, GPS-denied navigation), transfer to hardware is left to future work. *Reproducibility:* code, configurations, seeds, and logs are released publicly.

VI. CONCLUSION

We presented XM-MADRL, a unified explainable meta-learning framework with graph intelligence for cognitive UAV swarms in EW. Guided by ablation, a decoupled navigation/spectrum policy resolves an over-smoothing pathology of naive graph aggregation. Evaluated fairly at equal budget, XM-MADRL matches strong DRL baselines on mission success, energy, and safety while adding auditable SHAP explanations they lack; we report this parity transparently rather than claiming an unearned advantage. The lasting contributions are the unified interpretable architecture, the decoupled-head insight, and a fully reproducible benchmark and evaluation. Future work targets strongly partially-observable, non-stationary EW tasks—where coordination and meta-adaptation should translate into measurable gains—as well as hardware-in-the-loop validation.

REPRODUCIBILITY

All code, configurations, and result logs are available at <https://github.com/adeliusa486/XM-MADRL-eXplainable-Meta-learning-Multi-Agent-DRL>.

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